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COMPLEX BEND VACUUM CHAMBER FOR NSLS-II UPGRADE

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Abstract

While the NSLS-II synchrotron is a third-generation light source offering outstanding brightness and flux, a robust R&D program is underway to upgrade the facility to a fourth-generation, or beyond, capability. Inherent in the so-called complex-bend magnet and lattice designs are significant limitations on the beam channel and exit slot apertures of the vacuum chamber. These restrictions, along with the need for the vacuum chamber to be mechanically aligned and decoupled from the magnets, present unique challenges. For our chamber, the selected solution is not novel: it utilizes an aluminum split clamshell design, a proven approach used in many machines, past and present. The adaptation of this design, combined with precision machining and welding, is expected to provide the most cost-effective solution. Geometric and resistive-wall impedance considerations, along with structural and thermal modeling, will be presented, as well as dynamic pressure simulations generated using SynRad+ and MolFlow+ modeling codes. With ongoing changes in lattice and magnet parameters, a systematic, iterative approach to vacuum design has been implemented and will also be discussed.

INTRODUCTION

NSLS-II is based on the well-established double-bend achromat lattice (Fig. 1). Commissioned with first light in 2014, it was the last synchrotron constructed using this lattice type. Since then, new and upgraded synchrotrons have adopted various forms of multi-bend achromats to reduce horizontal emittance.

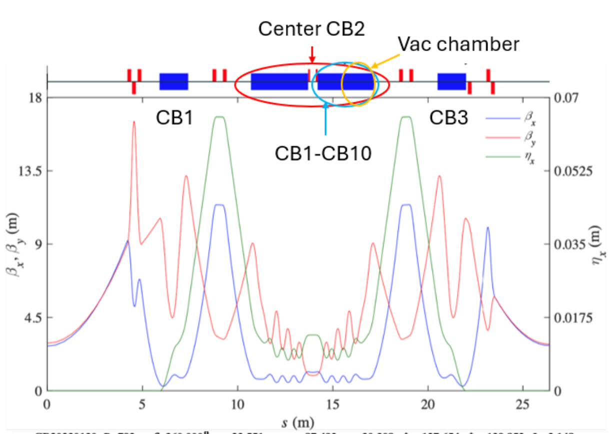

Figure 1: TCBA lattice showing central bending region CB2 of interest.

The goal of NSLS-II-U is to select a lattice with world-leading emittance and brightness. To achieve this, lattice designs based on the complex-bend principle are being investigated [1]. One promising candidate is the triple complex-bend achromat (TCBA), which incorporates three bending regions per cell [2]. The central bending region forms the basis of this prototype effort, as it imposes the most stringent constraints on magnet and vacuum chamber design due to its reduced aperture requirements.

MECHANICAL DESIGN

The vacuum chamber design was required to meet several key criteria: it needed to accommodate two different PMQ geometries (focusing and defocusing), maintain adequate clearance between the chamber and magnets along its entire length, and achieve an average base pressure below 1×10^{-9} Torr. Additionally, it was necessary to provide a smooth, continuous internal beam aperture of Ø11 mm, incorporate an exit slot for beam extraction, and structurally withstand the forces associated with vacuum conditions (Fig. 2).

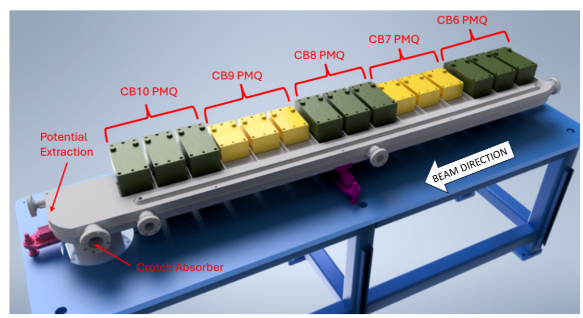

Figure 2: Arrangement of complex-bend PMQs with vacuum chamber (defocusing magnets shown in green, focusing magnets in yellow).

Based on the vertical emittance angle (Fig. 3), a vertical exit slot height of 3.5 mm was determined to be sufficient for extracting the bending magnet fan over a length of approximately 2 meters.

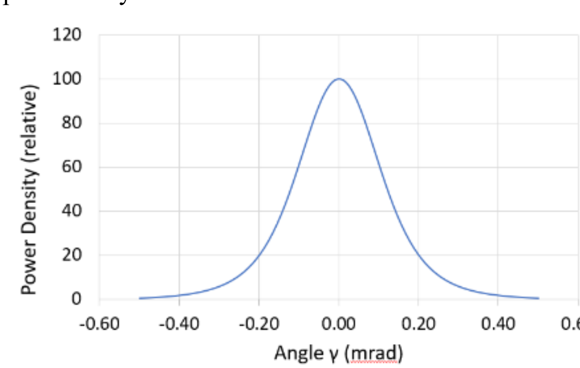

Figure 3: Approximate vertical dipole power density vs Angle.

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A key factor in the chamber design is the beam-induced impedance and its impact on both beam dynamics and chamber heating. We calculated the power generated by the beam's interaction with both geometric and resistive-wall impedance. Geometric impedance was simulated using GdfidL [3], taking into account parameters such as the beam channel profile, exit slot height and width, and ante-chamber height. The cross section was optimized to reduce low-frequency resonances, with the final geometry shown in Fig. 4.

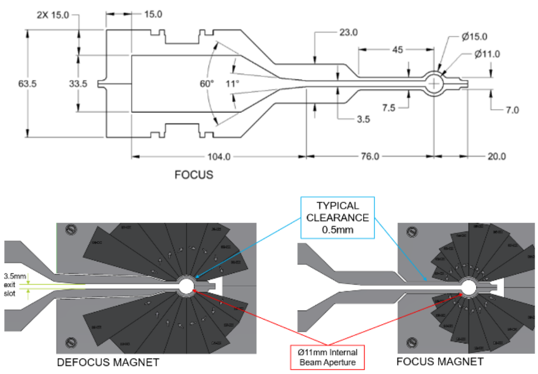

Figure 4: Cross section of optimized chamber.

Figure 5 shows the real part of the geometric impedance, the bunch profile (σ_s = 3 mm), and the normalized imaginary part of the geometric impedance for the prototype geometry. A bunch length of 3 mm represents the worst-case scenario. As shown, resonance peaks occur at frequencies above 15 GHz. The resulting power loss due to geometric impedance is minimal, estimated to be less than 1 W.

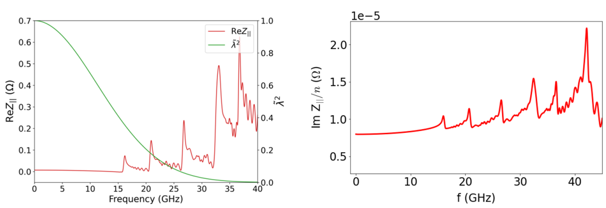

Figure 5: Real part of the geometric impedance and bunch profile (left) and normalized imaginary part of the geometric impedance for the prototype geometry (right).

The total beam-induced power as a function of bunch length is shown in Fig. 6. For a bunch length of 3 mm, the total power is approximately 13 W per meter.

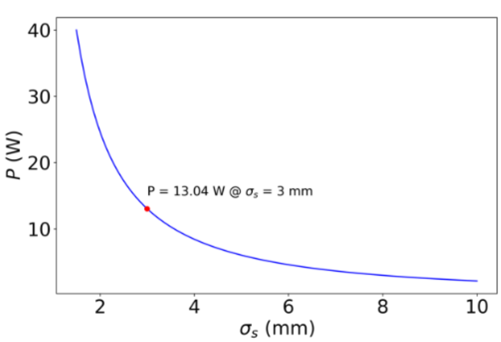

Figure 6: Total beam-induced power (geometric + resistive) as a function of bunch length.

The lattice information was used to generate a sweep of the chamber's internal beam channel along the precise nominal beam path, accounting for the different turning radii of the focusing and defocusing magnets. The outer beam channel of the vacuum chamber is kept straight within each PMQ to maintain clearance, with a small drift space and angle introduced between adjacent PMQs. The overall bend angle of the lattice (2.75°) was preserved (Fig. 7).

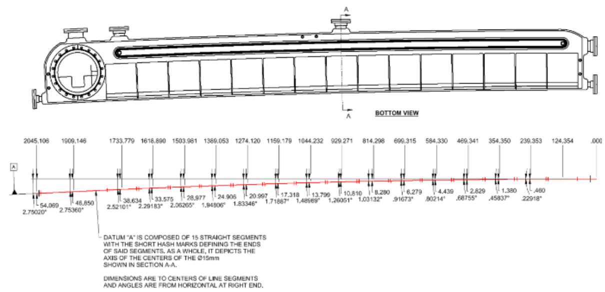

Figure 7: CAD model of the chamber sweep, illustrating the discrete angles and positions of the PMQs.

The lattice file was also used as input to SynRad+ [4]. SynRad+ confirmed that the generated beam path and the bending magnet fan can be successfully extracted through the exit slot for crotch absorber interception.

Furthermore, SynRad+ results showed negligible power deposition on the exit slot surfaces. The simulation reported a total power of 1114.87 W, with 1114.04 W on the end facet and only 0.4 W on the top and bottom exit slot surfaces (Fig. 8).

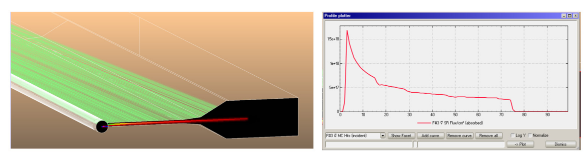

Figure 8: SynRad+ results of CB1-CB10 showing complete intercept of fan at end face.

With the overall length and apertures of the PMQs, the internal vacuum chamber cross section, and the beam curvature defined, refinements can now be made to the chamber design to optimize structural strength. Finite Element Analysis (FEA) reveals acceptable stress and deformation under vacuum (Fig. 9), with a maximum deflection of 0.10 mm per side and a maximum stress of 44 MPa.

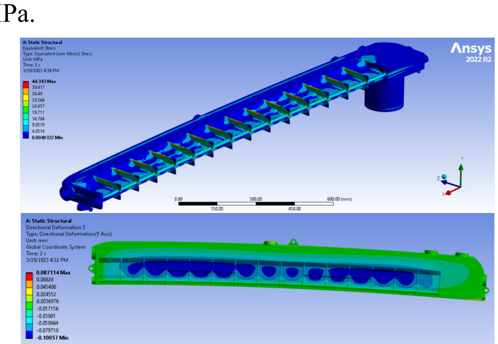

Figure 9: Structural analysis of the vacuum chamber with external ribs (no internal supports) under vacuum load. Top: Stress distribution. Bottom: Deformation profile.

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For the thermal analysis, a total heat load of 150 W was applied. Comprising 100 W on a single exit slot surface to simulate partial interception of the extracted synchrotron fan, and 50 W on the beam pipe to account for beam-induced power. With integrated cooling, the FEA results show a maximum temperature of 32 °C on the e-beam channel, well within safe limits for aluminum and unlikely to radiate significant heat to the surrounding PMQ magnets (Fig. 10).

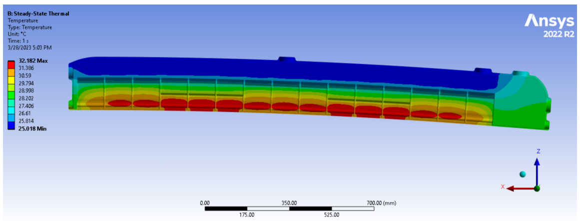

Figure 10: Thermal analysis showing temperature distribution with 50 W applied to the beam pipe and 100 W on the top exit slot (150 W total).

A vacuum level below $1{\times}10^{-9}$ Torr is required to minimize electron losses due to interactions with residual gas, thereby extending the beam lifetime well beyond the Touschek lifetime. Molflow+ simulations confirm that this vacuum level is achievable after 100 A·hr of beam dose [4] (Fig. 11).

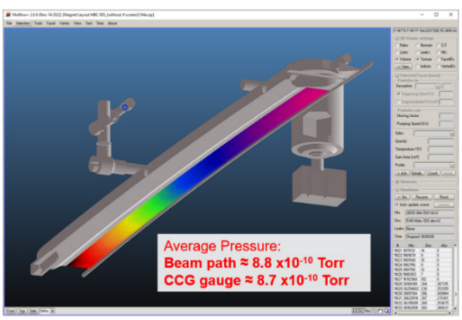

Figure 11: Molflow+ vacuum simulation.

The supports must accurately position the chamber while also allowing free movement during thermal bakeout. A specific arrangement of three sliding supports enables this functionality, holding the chamber rigid at room temperature while allowing expansion at elevated temperatures, functioning as a live support (Figs. 12 and 13).

CONCLUSION

The vacuum chamber was successfully fabricated, meeting all design criteria and maintaining an end-to-end flatness under 200 microns (Fig. 14). Thermal and structural analyses confirmed safe operating temperatures and acceptable stress levels. The three-point sliding support system allowed precise positioning and thermal expansion during bakeout. Vacuum simulations demonstrated that the required pressure below $1{\times}10^{-9}$ Torr is achievable. Overall, the chamber design and fabrication demonstrate readiness for operational conditions with reliable mechanical, thermal, and vacuum behavior.

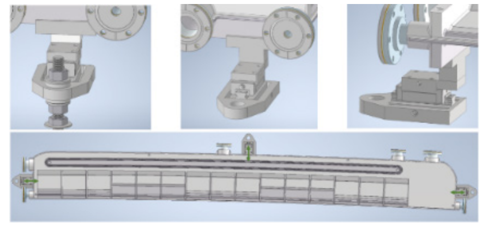

Figure 12: Chamber support and orientation of linear bearing movement.

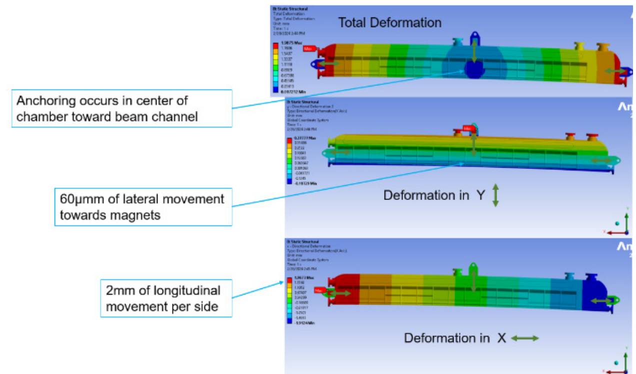

Figure 13: Thermal-structural analysis showing chamber movement at 100 °C.

Figure 14: Full-scale complex-bend vacuum chamber (top) and full assembly (bottom).

ACKNOWLEDGMENTS

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The research described herein is Fundamental Research as defined in the ITAR (22 CFR §120.34(a)(8)), EAR (15 CFR §734.8), or Part 810 (10 CFR §810.3), as applicable, and as described in the USD (AT&L) memoranda on Fundamental Research, dated May 24, 2010, and on Contracted Fundamental Research, dated June 26, 2008.

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